DNS SERVER CONFIGURATION ASSIGNMENT

1. Install and set up a DNS server using BIND on Linux or Simple DNS Plus on Windows, then create a DNS zone for musicmania.com.

Setting	Values
DNS server hostname	dns1.musicmania.com
DNS server IP address	127.0.0.53
DNS zone name	musicmania.com
Zone type	Master / Primary

Steps

1. Install BIND (for example, sudo dnf install bind bind-utils on Fedora).
2. Add a zone declaration for musicmania.in in /etc/named.conf.
3. Create a zone file under /var/named/ containing the DNS records.
4. Validate the configuration and zone file, then enable and start the named service.

zone declaration

```
zone "musicmania.com" IN {
   type master;
   file "musicmania.com.zone";
};
```

zone file: musicmania.com.zone

```
$TTL 86400
@   IN  SOA dns1.musicmania.com. admin.musicmania.com. (
        2026111001 ; Serial
        3600       ; Refresh
        900        ; Retry
        604800     ; Expire
        86400 )    ; Minimum


@       IN  NS  dns1.musicmania.com.
dns1    IN  A   127.0.0.53
```

Explanation: The zone file defines the zone and its DNS records. The NS record identifies the name server, while A records map hostnames to IPv4 addresses.

2. Add an A record mapping www.musicmania.com to 192.168.1.50, then use nslookup or dig to verify the mapping.

A record to add: www.musicmania.com IN A 127.0.0.53

Verification commands

```
nslookup www.musicmania.com 127.0.0.53
dig @192.168.1.10 www.musicmania.com A
```

nslookup www.musicmania.com
Server: 127.0.0.53
Address: 127.0.0.53#53

Non-authoritative answer:
Name: www.musicmania.com
Address: 139.162.174.209
Name: www.musicmania.com
Address: 172.104.251.198
Name: www.musicmania.com
Address: 172.104.203.186
Name: www.musicmania.com
Address: 172.104.149.86
Name: www.musicmania.com
Address: 139.162.181.76
Name: www.musicmania.com
Address: 2a01:7e01::2000:dff:fec1:5cdb
Name: www.musicmania.com
Address: 2a01:7e01::f03c:95ff:feed:783e
Name: www.musicmania.com
Address: 2a01:7e01::f03c:95ff:fe00:2692
Name: www.musicmania.com
Address: 2a01:7e01::f03c:95ff:fe91:5d91
Name: www.musicmania.com
Address: 2a01:7e01::f03c:95ff:feeb:9d4b

3. Add a CNAME record for play.musicmania.com pointing to www.musicmania.com, then test whether it resolves to the correct IP address.

CNAME record to add: play.musicmania.com. IN CNAME www.musicmania.com.

Explanation: A CNAME record makes one hostname an alias of another. DNS follows the alias to www.musicmania.com and returns the IP address associated with that name.

4. The DNS server is not resolving musicmania.com as expected. Document a step-by-step troubleshooting process, including configuration checks, restarting the DNS service, and testing with nslookup or dig.

Step	Command / check	Purpose
1. Check configuration	sudo named-checkconf	Checks BIND configuration syntax.
2. Check zone file	sudo named-checkzone musicmania.com /var/named/musicmania.com.zone	Checks the zone file for errors.
3. Restart service	sudo systemctl restart named	Restarts BIND to apply changes.

4. Check service status	sudo systemctl status named	Confirms whether the DNS service is running.
5. Test DNS resolution	dig @127.0.0.53 www.musicmania.com A	Queries the local DNS server and checks the A record.

Additional check: Confirm the client is querying the correct DNS server (127.0.0.53), that the server is reachable, and that firewall rules allow DNS traffic on UDP and TCP port 53.

Conclusion: DNS problems may be caused by incorrect records, configuration syntax errors, a stopped DNS service, or clients using the wrong DNS server. Configuration checks, a service restart, and dig/nslookup tests help locate the issue.